

Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term

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March 3, 2022

Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation $\partial u / \partial t - \frac{1}{2} \Delta u = b(u) \dot{W}$ for a Lipschitz continuous diffusion coefficient b which is not necessarily bounded. Using the theory of the stochastic integral laid out by John Walsh, we provide conditions which will guarantee the existence of an invariant measure for a broad range of initial conditions which includes L^2_ρ as well as the Dirac delta distribution δ_0 .

MSC 2010 subject classification: 60H15, 60H07, 60F05.

Keywords: Stochastic heat equation, parabolic Anderson model, invariant measure, delta initial condition.

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1 Introduction

We consider the following stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot), \end{cases} \quad (1.1) \quad \boxed{\text{E:gspde}}$$

where $b(x, u)$ is assumed to be a globally Lipschitz continuous function in u with Lipschitz constant independent of x . In particular, our assumption allows the linear case $b(x, u) = \lambda u$, which is usually referred to as the *parabolic Anderson model* [CM94]. In this equation, $\dot{W}(t, s)$ is a centered Gaussian noise that is white in time and homogeneously correlated in space. The initial condition μ is a deterministic, locally finite, regular, signed Borel measure that satisfies (1.3) below. The aim of the paper is to investigate the conditions required to guarantee the existence of invariant measures for the solution to (1.1), which is a crucial step towards the study of the ergodicity of the system.

The question of the existence of invariant measures in unbounded domains has been studied in few papers such as [AM03, MSY15, MSY16, TZ98]. Since the spatial domain is unbounded, one needs to introduce a weight function ρ and the corresponding weighted space $L^2_\rho(\mathbb{R}^d)$ as usual; see Section 2.3 below for more details. Following the standard procedure as laid out by Da Prato and Zabczyk [DZ92b], the main effort is to show that

$$\sup_{t>0} \mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) < \infty. \quad (1.2) \quad \text{E:BddMnt}$$

Recall that under the following Assumption 1.1, one can find a unique mild solution to (1.1) with sharp moment estimates; see [CK19, CH19, CD15].

Assumption 1.1. (i) The initial measure μ satisfies the following integrability condition:

$$\int_{\mathbb{R}^d} \exp(-a|x|^2) |\mu|(dx) < \infty \quad \text{for any } a > 0; \quad (1.3) \quad \text{A:icon}$$

(ii) The spectral measure $\hat{f}$ satisfies *Dalang's condition*:

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty \quad \text{for some and therefore any } \beta > 0; \quad (1.4) \quad \text{A:dalcon}$$

(iii) The diffusion coefficient $b(x, u)$ satisfies the following Lipschitz and boundedness conditions:

$$|b(x, u) - b(x, v)| < L_b |u - v| \quad \text{and} \quad |b(x, 0)| \leq L_0 \quad \text{for all } u, v \in \mathbb{R}, x \in \mathbb{R}^d. \quad (1.5) \quad \text{A:lipcon}$$

Moreover, this solution is usually *intermittent*, namely, the probability moments to (1.1) have exponential growth in t . In order to have (1.2), one has to first identify the exact conditions, under which the moments to (1.1) do not possess exponential growth. This question has been answered in [CK19], where necessary and sufficient conditions are given. In [AM03, MSY15, MSY16], a drift term $f(x, u)$ is included in (1.1) in a crucial way to help to cancel the otherwise exponential growing moments. The absence of such a drift term in this paper makes the problem more challenging. According to part (3) of Theorem 1.3 of [CK19], in order to have (1.2), one has to have L_b in (1.5) — the Lyapunov exponent of b — small enough and

$$\Upsilon(0) < \infty, \quad (1.6) \quad \text{E:U0finite}$$

where $\Upsilon(\cdot)$ is defined in (1.4). Note that condition (1.6) is equivalent to

$$d \geq 3 \quad \text{and} \quad \int_{\mathbb{R}^d} \frac{1}{|z|^{d-2}} \Gamma(dz) < \infty, \quad (1.7) \quad \text{E:U0Eqv}$$

where Γ is the correlation measure (see (2.1)). Condition (1.6) (or equivalently (1.7)) gives the admissible correlation functions, which is weaker than that in [TZ98]; see Remark 1.2 below.

R:Cor

Remark 1.2. The assumption on the correlation function in [TZ98] is given through the spectral density $\widehat{f}$ as follows:

$$\widehat{f} \in L^p(\mathbb{R}^d) \quad \text{for some } p \in (d/(d-2), \infty]; \quad (1.8)$$

E:CorTZ

see Hypothesis 2.1 (*ibid.*) and recall that $d \geq 3$. One can easily check if $\widehat{f}$ satisfies (1.6), then one can find a p in the range given in (1.8) such that condition (1.8) is satisfied. In some sense, our condition (1.6) seems to be more precise and convenient to check.

Nick, can you check the relation between the two conditions: (1.6) and (1.8)?

I constructed the example in the discussion document that shows that neither our condition nor their condition is more general than one another. We can find $\widehat{f}$ that satisfies Dalang's condition but not Hypothesis 2.1 [TZ98] and vice versa. We never completely verified this but if true then we should make a comment about this.

Here is the main theorem of the paper:

T:Main

Theorem 1.3. *Under the following conditions:*

(i) Condition (1.6) (or equivalently (1.7)) holds and for some $\alpha \in (0, 1/2)$,

$$\int_0^t dr \, r^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-r|\xi|^2) < \infty, \quad \text{for all } t > 0; \quad (1.9)$$

A:fcon

(ii) Suppose that for any admissible weight ρ , there exists another admissible weight $\widehat{\rho}$ with

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty \quad (1.10)$$

E:rho

such that the initial condition satisfies the following condition:

$$\limsup_{t>0} \int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} G(t, x-y) |\mu|(dy) \right]^2 \widehat{\rho}(x) dx < \infty; \quad (1.11)$$

E:InitData

Considering that we still need to assume the admissible weight property because if its use in Proposition 2.1 [TZ98], I don't think that this condition helps us. I think we should stick with the assumption that for all $t > 0$, there exists a $K(t) \in \mathbb{N}$ such that

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |\mu|)(x) < K(t).$$

(iii) The Lipschitz constant L_b of b (see (1.5)) satisfies

$$L_b < \dots, \quad (1.12)$$

E:LipLb

we have that

(a) $u(t, \cdot) \in L^2_{\widehat{\rho}}(\mathbb{R}^d)$ for all $t > 0$ with (1.2) satisfied;

(b) there exists an invariant measure for the laws of $\{u(t, \cdot)\}_{t>0}$.

Statement (b) needs to be for $t > t_0$. A measure η is invariant for the laws $\{\mathcal{L}(u^a(t, \cdot))(A)\}_{t>0}$, $A \in \mathcal{B}(L^2_\rho(\mathbb{R}^d))$ and $a \in L^2_\rho(\mathbb{R}^d)$, if

$$\eta(A) = \int_{\mathcal{B}(L^2_\rho(\mathbb{R}^d))} \mathcal{L}(u^a(t, \cdot))(A) \eta(da) \quad \text{for all } t > 0.$$

However we can only show that there exists an measure, η , that is invariant for $\{\mathcal{L}(u^a(t, \cdot))(A)\}_{t>t_0}$ for $t_0 > 0$ arbitrarily small. In other words, what we were able to show was

$$\eta(A) = \int_{\mathcal{B}(L^2_\rho(\mathbb{R}^d))} \mathcal{L}(u^a(t, \cdot))(A) \eta(da) \quad \text{for all } t > t_0.$$

One thing to note is that the just mentioned papers use the theory of the stochastic integral laid out by Da Prato and Zabczyk in [DZ92a] while in this paper we use the theory developed by Walsh in [Wal86]. The former develops a theory of integration with respect to Hilbert-space-valued processes while the later emphasizes integration with respect to worthy martingale measures. Both theories have their advantages and disadvantages but have been proven to be more or less equivalent in [DQ11].

The need to prove existence of invariant measures is apparent for the fact that they are crucial in order to study the ergodicity of dynamical systems. Da Prato and Zabczyk [DZ92b] established the following procedure

- Show the existence of a Markovian solution to the SPDE in question in a certain function space where the transition group satisfies the Feller Property
- Show that the solution starting from a particular initial condition is bounded in probability

The first bullet point has been shown to be true [DZ96] for a wider class of SPDEs which includes (1.1) and in fact has also been shown to be true in other various cases of interest [Cer01, Mas99]. In [TZ98] they follow this procedure to show the existence of an invariant measure for the solutions to (1.1) in two separate cases:

1. The initial measure is a function, $\varphi \in L^2_\rho \cap L^2_{\hat{\rho}}$ where both ρ and $\hat{\rho}$ satisfy the condition that $\rho \cdot (\hat{\rho})^{-1} \in L^1(\mathbb{R}^d)$ and the solution is bounded in probability.
2. The initial measure is a constant function while assuming certain integrability properties of the spectral density.

By now using the theory of Walsh as well as subsequent results proven in [CH19] we are able to mimic the proof in [TZ98], but in the language of Walsh, and show the existence of invariant measures for a more broad group of initial conditions. However, because of the differences between the two theories of the stochastic integral, we must trade some of previous assumptions on the spectral density for new ones which are listed below. For example we drop hypothesis 2.1(i) as well as the assumptions from [TZ98, Theorem 3.3] for the new assumptions listed below.

Misiats, Stanzhytski and Yip [MSY15] show the existence of an invariant measure for (1.1) with any $L^2(\mathbb{R}^d)$ initial condition but with non-zero drift term $f(x, u(t, x))$ added to the right side of the equal sign. However their drift term f and diffusion coefficient b must be bounded above by a some $\varphi \in L^\infty(\mathbb{R}^d) \cap L^1(\mathbb{R}^d)$ in the sense that

1. $|f(x, 0)|, |b(x, 0)| \leq \varphi(x)$
2. $|f(x, u) - f(x, v)|, |b(x, u) - b(x, v)| \leq L\varphi(x)|u - v|$

In our paper we do not require such an upper bound. We also allow for any $L^2_\rho(\mathbb{R}^d)$ initial condition which contains the $L^2(\mathbb{R}^d)$ initial conditions. However we only consider the case where the drift term $f \equiv 0$. For the case of unbounded coefficients, Manthey and Assing [AM03] have showed the existence of an invariant measure when considering the weight $\rho(x) = (1 + |x|^2)^{-a/2}$, $a > d$ and a bounded continuous initial condition. Their diffusion coefficient follows the same growth properties as ours but they allow for non-zero drift term that follows a "one-sided" linear growth condition. In their paper they are able to squeeze the solution to their SPDE in between two other solutions, each satisfying a desired inequality that leads to the existence of an invariant measure. As of now, the theory of Walsh can not be applied to such drift coefficients so it is unclear if the results in this paper may apply to such a SPDE.

Before we state the main results of this paper we need to propose series of assumptions. First, we have the following:

With Assumption 1.1, we can now state two already established results. The first result deals with the existence and uniqueness of the solution of (1.1). The second result gives a bound on the moments p^{th} -moments of the solution, where we denote $\|X\|_p := (\mathbb{E}|X|^p)^{1/p}$ to be the p^{th} moment of a random variable X . It should be noted that that proof of our main result "Insert main result" heavily relies on the moment bound given in Theorem 1.5 below.

Theorem 1.4 (Theorem 1.2 [CH19]). *Under Assumption 1.1, the SPDE (1.1) has a unique random field solution starting from μ and the solution is $L^2(\Omega)$ continuous.*

T:ExUn

Theorem 1.5 (Theorem 1.7 [CH19]). *Under Assumption 1.1, for any $t > 0$ and $x \in \mathbb{R}^d$, the solution to (1.1), $u(t, x)$, given by (2.2) is in $L^p(\Omega)$ and for $p \geq 2$*

$$\|u(t, x)\|_p \leq [\bar{\varsigma} + \sqrt{2}(|\mu| * G(t, \cdot))(x)] H(t; \gamma_p)^{1/2}, \quad (1.13) \quad \text{E:mb}$$

where $\bar{\varsigma} = L_0/L$, $\gamma_p = 32pL^2$ and the function $H(t; \gamma_p)$ is nondecreasing in t (see [CH19] for the concrete definition of H). Moreover, if we assume that

$$\Upsilon(0) < \infty \quad (1.14) \quad \text{A:dalcon0}$$

then this function $H(t; \gamma_p)$ is bounded in t . Hence, by choosing $C = (\sqrt{2} \vee \bar{\varsigma}) \sup_{t \geq 0} H(t; \gamma_p)^{1/2} < \infty$, we have that

$$\|u(t, x)\|_p \leq \begin{cases} C [1 + (|\mu| * G(t, \cdot))(x)] & \text{if } L_0 > 0, \\ C (|\mu| * G(t, \cdot))(x) & \text{if } L_0 = 0. \end{cases} \quad (1.15) \quad \text{E:mb0}$$

1.1 Main Results

Our first main result handles the case of the existence of an invariant measure for $L_\rho^2(\mathbb{R}^d)$ initial conditions (see Section 2.3), namely the following SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, x) = \varphi(x) & \varphi(x) \in L_\rho^2(\mathbb{R}^d) \end{cases}. \quad (1.16) \quad \text{E:spde}$$

Before we state the result, we need to make an additional assumption on the spectral measure.

With this assumption now stated, we now state our first main result, Theorem 1.6.

T:Main

Theorem 1.6. *Assume that Assumption 1.1 holds with the stronger assumption that $\Upsilon(0) < \infty$. Also assume (1.9). Now for any admissible weight ρ , suppose there exists another admissible weight $\hat{\rho}$ and an element $\hat{\varphi} \in L_\rho^2(\mathbb{R}^d) \cap L_{\hat{\rho}}^2(\mathbb{R}^d)$ such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} dx < \infty$$

and $u^{\hat{\varphi}}(\cdot, x)$ is bounded in probability in $L_\rho^2(\mathbb{R}^d)$. Then for any $t_0 > 0$ there exists an invariant measure for the laws of the solution of the SPDE (1.1), $\{\mathcal{L}(u^{\hat{\varphi}}(t, \cdot))\}_{t \geq t_0}$, in $L_\rho^2(\mathbb{R}^d)$ for any $L_\rho^2(\mathbb{R}^d)$ initial condition.

We now state our last assumption which will be required for the proof of one of our last main result, Theorem (1.9).

Assumption 1.7.

$$\sup_{x \in \mathbb{R}^d} (G(t, \cdot) * |\mu|)(x) < K(t), \quad \forall t > 0 \text{ and } K(t) \in \mathbb{N} \quad (1.17) \quad \text{A:newgicon}$$

Remark 1.8. It is easy to see that Assumption (1.17) is satisfied whenever $\mu = \varphi \in L^2_\rho(\mathbb{R}^d)$ is bounded and also when $\mu = \delta_0$ where δ_0 is the Dirac delta measure.

T:Main2

Theorem 1.9. Suppose the assumptions of Theorem 1.6 remain true and in addition assume (1.17) holds true. Suppose there existed another admissible weight $\hat{\rho}$ such that

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\hat{\rho}(x)} dx < \infty.$$

Then there exists an invariant measure for the laws of the solutions to (1.1) in $L^2_\rho(\mathbb{R}^d)$ for $t > l > 0$. In particular, for any $l > 0$ there exists a measure that is invariant for $\{\mathcal{L}(u^\mu(t, \cdot))\}_{t>l}$ where u is the solution to (1.1) with initial measure μ .

C:delta

Corollary 1.10. Suppose the assumption of Theorem 1.6 remain true. Then for any $l > 0$, the stochastic heat equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \delta_0(\cdot) \end{cases} \quad (1.18) \quad \text{E:dspde}$$

has an invariant measure in L^2_ρ for the laws of their solution for $t > l$ and for any ρ which there exists $\hat{\rho}$ such that $\rho/\hat{\rho} \in L^1(\mathbb{R}^d)$.

Proof. The SPDE above satisfies (1.17). By Lemma 3.4 we know that for an arbitrary $t_0 > 0$, the solution at $t = t_0$ from the SPDE is in $L^2_\rho \cap L^2_{\hat{\rho}}$. Theorem 1.9 implies the result. $\square$

2 Preliminaries

2.1 Definition of the Gaussian noise and Walsh integral

Following [DK⁺09], a Gaussian noise that is white in time and colored in space is an $L^2(\Omega, \mathcal{F}, \mathbb{P})$ -valued zero mean Gaussian Processes on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$

$$\{F(\psi) : \psi \in C_0^\infty(\mathbb{R}^{d+1})\},$$

such that

$$\mathbb{E}[F(\psi) F(\varphi)] = J(\psi, \varphi) := \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\psi(s, \cdot) * \tilde{\varphi}(s, \cdot))(x) \quad (2.1) \quad \text{E:Cor}$$

where $\tilde{\varphi}(x) := \varphi(-x)$ and Γ is a tempered measure on $\mathbb{R}^d$ that is nonnegative and nonnegative definite. When Γ has density $\Gamma(dx) = f(x)dx$ then J above can be written as

$$J(\psi, \phi) = \int_0^\infty ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy \psi(s, x) f(x - y) \phi(s, y).$$

f is often referred to as the *spatial correlation* function of the Gaussian Noise and its Fourier transform

$$\hat{f}(\xi) = \mathcal{F}f(\xi) = \int_{\mathbb{R}^d} \exp(-i\xi \cdot x) f(x) dx$$

is often referred to as the spectral density. Now following [Wal86] and [DK⁺09] and by denoting $\mathcal{B}_b(\mathbb{R}^d)$ as the family of bounded Borel subsets of $\mathbb{R}^d$, one may construct out of this a worthy martingale measure $W = (W_t(A), t \geq 0, A \in \mathcal{B}_b(\mathbb{R}^d))$ where

$$W_t(A) := \lim_{t \rightarrow \infty} F(\psi_n),$$

where the limit is in $L^2(\Omega, \mathcal{F}, \mathbb{P})$, $\psi_n \in C_0^\infty(\mathbb{R}^{d+1})$ and $\psi_n \downarrow 1_{[0,t] \times A}$. With this construction, one may show that $(W_t(A), t \geq 0, A \in \mathcal{B}_b(\mathbb{R}^d))$ is a worth martingale measure and its covariation measure Q is given by

$$Q(A \times B \times [s, t)) = (t - s) \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy 1_A(x) f(x - y) 1_B(y)$$

and its dominating measure is $K \equiv Q$. The main point is that F and W are related by

$$F(\psi) = \int_0^\infty \int_{\mathbb{R}^d} \psi(t, x) W(dt, dx)$$

where the integral is a martingale measure stochastic integral as in Walsh with filtration given by

$$\mathcal{F}_t = \sigma(W_s(A), s \leq t, A \in \mathcal{B}_b(\mathbb{R}^d)) \vee \mathcal{N}$$

where $\mathcal{N}$ is the sigma field generated by the $\mathbb{P}$ -null sets.

Now for any adapted, jointly measurable processes (with respect to $\mathcal{B}((0, \infty) \times \mathbb{R}^d) \times \mathcal{F}$) random field $\{X(t, x) : t > 0 \text{ and } x \in \mathbb{R}^d\}$ such that the for all integers $p \geq 2$,

$$\int_0^\infty ds \iint_{\mathbb{R}^{2d}} dx dy \|X(s, y) X(s, x)\|_{\frac{p}{2}} f(x - y) < \infty,$$

the stochastic integral

$$\int_0^\infty \int_{\mathbb{R}^d} X(s, y) W(ds, dy)$$

is well defined in the sense of Walsh. Throughout this paper $\|\cdot\|_p$ will denote the standard $L^p(\Omega)$ norm, $(\mathbb{E}|\cdot|^p)^{\frac{1}{p}}$.

With all this said, we define the mild solution to (1.1) as

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy) \quad (2.2) \quad \boxed{\text{E:gmsol}}$$

where the and integral above is a Walsh integral and

$$G(t, x) = (2\pi t)^{-d/2} \exp\left(-\frac{|x|^2}{2t}\right).$$

We also denote

$$J(t, x) := \int_{\mathbb{R}^d} G(t, x - y) \mu(dy)$$

with the distinction that

$$J_0(t, x) := \int_{\mathbb{R}^d} G(t, x - y) |\mu|(dy)$$

and lastly,

$$I(t, x) := \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy).$$

D:Sol

Definition 2.1. A process $u = (u(t, x), (t, x) \in (0, \infty) \times \mathbb{R}^d)$ is called a random field solution to (1.1) if:

1. u satisfies almost surely and for all $(t, x) \in (0, \infty) \times \mathbb{R}^d$

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) + \int_0^t \int_{\mathbb{R}^d} G(t - s, x - y) b(y, u(s, y)) W(ds, dy);$$

2. u is adapted, that is, for all $(t, x) \in (0, \infty) \times \mathbb{R}^d$, $u(t, x)$ is $\mathcal{F}_t$ measurable;
3. u is jointly measurable with respect to $\mathcal{B}((0, \infty) \times \mathbb{R}^d) \times \mathcal{F}$;
4. $\|I(t, x)\|_2 < +\infty$ for all $(t, x) \in (0, \infty) \times \mathbb{R}^d$;
5. I is $L^2(\Omega)$ -continuous, that is, the function $(t, x) \mapsto I(t, x)$ mapping $(0, \infty) \times \mathbb{R}^d$ into $L^2(\Omega)$ is continuous.

2.2 Invariant measures

Let $(E, \mathcal{E})$ denote an arbitrary measure space and $\mathcal{M}_1(E)$ be the set of all probability measures on $(E, \mathcal{E})$. Let $P_t(x, A)$, $t \geq 0$, $x \in E$, $A \in \mathcal{E}$ be a Markovian Transition function as in [DZ96]. Associated with each Markovian transition function is a semigroup of linear operators, P_t defined on the set of bounded measurable functions on E , $\mathcal{B}_b(E)$ defined by

$$P_t \varphi(x) = \int_E P_t(x, dy) \varphi(y).$$

Lastly we say that the above semigroup of linear operators satisfies the Feller Property if for each continuous and bounded function φ on E we have that $P_t \varphi(x)$ is also a bounded and continuous function on E . Now consider, for any $\mu(\cdot) \in \mathcal{M}_H$, the laws of the solution $u(t, x)$ of (1.1)

$$\mathcal{L}(u^\mu(t, \cdot))(A) = \mathbb{P} \left\{ \omega \in \Omega : u^\mu(t, \cdot) \in A \mid A \in \mathcal{B}(L_\rho^2(\mathbb{R}^d)) \right\}$$

and define $P_t(\mu, A) := \mathcal{L}(u^\mu(t, \cdot))(A)$. The following two definitions will define what it means for a measure to be invariant for an SPDE.

Definition 2.2. A measure $\eta \in \mathcal{M}_1(E)$ is *invariant* for P_t if

$$\eta(A) = \int_E P_t(x, A) \eta(dx) \quad \text{for each } t \geq 0 \text{ and } A \in \mathcal{E}.$$

Definition 2.3. A measure η is *invariant for the SPDE (1.1)* if η is invariant for $P_t(\mu, A) := \mathcal{L}(u^\mu(t, \cdot))(A)$.

Remark 2.4. Lemma 2.7 below shows that $L_\rho^2(\mathbb{R}^d) \subset \mathcal{M}_B$.

As mentioned in the introduction, we said our first goal was to establish the Feller property. However, this has been established in [DZ96] and in fact they verify this for a larger class of SPDEs. The second step was to show the solution is bounded in probability which means

$$\forall \epsilon > 0, \exists R > 0 : \forall t > 0, \mathbb{P} [|u^\varphi(t, \cdot)|_\rho \geq R] < \epsilon.$$

We verify this property and deduce the existence of an invariant measure for L_ρ^2 initial conditions in Theorem 1.6 and then generalize the result in Theorem 1.9 for measure valued initial conditions, and in particular, the Dirac delta measure δ_0 .

2.3 The weighted space $L^2_\rho(\mathbb{R}^d)$

Let ρ be some nonnegative function on $\mathbb{R}^d$. We denote by $L^2_\rho(\mathbb{R}^d)$ the weighted Hilbert space with the inner product $\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x)dx$ and set $\|f\|_\rho^2 := \int_{\mathbb{R}^d} f(x)^2\rho(x)dx$. Following Tessitore and Zabczyk [TZ98], we assume that ρ is an *admissible weight* meaning that ρ is a positive, bounded and continuous function in $L^1(\mathbb{R}^d)$ such that for $T > 0$ there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T)\rho(x) \quad \text{for all } t \in [0, T]. \quad (2.3) \quad \text{E:aw}$$

Several examples of admissible weights are (see Example in Section 2 of [TZ98])

$$\begin{aligned} \rho(x) &= \exp(-a|x|), & a > 0 \quad \text{and} \\ \rho(x) &= (1 + |x|^a)^{-1}, & a > d. \end{aligned} \quad (2.4) \quad \text{E:admRho}$$

R:admRho

Remark 2.5. Certainly, the smaller weight function $\rho(\cdot)$, the larger the space $L^2_\rho(\mathbb{R}^d)$ is. For example, one may choose ρ to be either a nonnegative function with compact support or the heat kernel itself $G(1, \cdot)$. In both cases, ρ is smaller than those in (2.4) (up to a constant). However, one can easily check that the admissible condition (2.3) excludes these two cases. It may be much less obvious that the following function

$$\rho(x) = \exp(-a|x|^b), \quad x \in \mathbb{R}^d$$

with $a > 0$ and $b \in (1, 2)$ is not admissible (Nick, can you prove this fact in appendix). In any case, we will focus on admissible weight functions as given in (2.4).

For the remainder of this section, we focus ourselves on (1.1) with initial condition $\varphi \in L^2_\rho(\mathbb{R}^d)$, namely the following stochastic heat equation.

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta u(t, x) = b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, x) = \varphi(x) & \varphi(x) \in L^2_\rho(\mathbb{R}^d) \end{cases} \quad (2.5) \quad \text{E:spde}$$

which has mild solution given by

$$u(t, x) = \int_{\mathbb{R}^d} G(t, x - y)\varphi(y)dy + \int_0^t \int_{\mathbb{R}^d} b(u(s, y))G(t - s, x - y)W(ds, dy). \quad (2.6) \quad \text{E:msol}$$

We will show that the solution, denoted as $u^\varphi(t, x)$, is an object in $L^2_\rho(\mathbb{R}^d)$ for each $t > 0$. We may denote the solution to (2.5) as $u(t, x)$ or simply as u when there is no confusion. First we state and prove a simple lemma showing that $\varphi \in L^2_\rho$ implies condition (1.3).

L:RhoDom

Lemma 2.6. *Let ρ be an admissible weight. Then we have the following inequality:*

$$\exp(-a|x|^2) \leq K_a \rho(x), \quad \text{for all } x \in \mathbb{R}^d. \quad (2.7) \quad \text{E:RhoDom}$$

Proof. The admissible weight property (2.3) tells us that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq C_\rho(T), \quad \forall t \in [0, T]. \quad (2.8) \quad \text{E:awbd1}$$

We note that in order to prove (2.7), it suffices to show that there exists some $K'_a \in \mathbb{N}$ such that

$$\lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq K'_a. \quad (2.9) \quad \text{E:awbd2}$$

Indeed, this would imply that there would exist an R_ϵ such that

$$\exp(-a|x|^2) \leq (K'_a + \epsilon)\rho(x) \quad |x| > R_\epsilon$$

and by the positivity and continuity of both $\exp(-a|x|^2)$ and $\rho(x)$, we know that there exists some positive constant $K'(a, \epsilon)$ such that

$$\exp(-a|x|^2) \leq K'(a, \epsilon)\rho(x) \quad |x| \leq R_\epsilon$$

and by then setting $K_a := K(a, \epsilon) = \max\{K'(a, \epsilon), (K'_a + \epsilon)\}$ then

$$\exp(-a|x|^2) \leq K_a\rho(x) \quad x \in \mathbb{R}^d$$

which is precisely (2.7).

In order to show (2.9), it suffices to show that

$$\lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\exp(-a|x|^2)} = \infty \quad (2.10)$$

E:awbd3

because this would mean that $\exp(-a|x|^2)$ is much smaller than $(G(t, \cdot) * \rho(\cdot))(x)$ for all large $|x|$ and since (2.8) is true then this would mean that (2.9) would be true as well by squeeze theorem,

$$0 \leq \lim_{|x| \rightarrow \infty} \frac{\exp(-a|x|^2)}{\rho(x)} \leq \lim_{|x| \rightarrow \infty} \frac{(G(t, \cdot) * \rho(\cdot))(x)}{\rho(x)} \leq C_\rho(T)$$

We prove (2.10) as follows: first recall $|\rho(x)| < M$ for some $M \in \mathbb{N}$. Then,

$$\begin{aligned} 0 \leq \frac{\int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) \rho(y) dy}{\exp(-a|x|^2)} &\leq \frac{M \int_{\mathbb{R}^d} \exp\left(-\frac{|x-y|^2}{2t}\right) dy}{\exp(-a|x|^2)} \\ &= \frac{M * \sqrt{2\pi t}}{\exp(-a|x|^2)} \\ &\rightarrow \infty \quad \text{as } |x| \rightarrow \infty \end{aligned}$$

Thus (2.10) is proved which means that (2.9) is also true.

No, you need to that a lower bound of the ratio goes to infinity.

□

L:phi

Lemma 2.7. Suppose that $\varphi \in L^2_\rho(\mathbb{R}^d)$. Then φ satisfies (1.3).

Proof. By Lemma 2.7 and the Cauchy-Schwartz inequality, there exists a constant K such that

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx \leq K \left(\int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)|^2 dx \right)^{1/2} \leq K \|\varphi\|_\rho < \infty.$$

□

Lemma 2.7 and Theorem 1.5 implies that (2.5) has a unique solution. The following theorem tells us that the solution for any $t > 0$, $u^\varphi(t, \cdot)$, is a element in $L^2_\rho(\mathbb{R}^d)$.

T: Moment

Theorem 2.8. Suppose Assumption 1.1 holds. Let $u(t, x)$ be the unique solution to (2.5) starting from $\varphi(\cdot) \in L^2_\rho$. Then for all $T > 0$ and for all $t \in [0, T]$, $u(t, \cdot) \in L^2_\rho(\mathbb{R}^d)$ with

$$\mathbb{E}(|u(t, \cdot)|^2_\rho) \leq 2C_T \left[\|\rho\|_{L^1(\mathbb{R}^d)} + C_\rho(T) |\varphi|_\rho^2 \right] < \infty. \quad (2.11)$$

E:uNormRho

where we use $C_T := H(T, \gamma_2)$. Moreover, if $\Upsilon(0) < \infty$ and assume (1.17), in other words, for each $t > 0$ there exists a $K(t) \in \mathbb{N}$ such that $\sup_{x \in \mathbb{R}^d} (|\varphi| * G(t, \cdot))(x) < K(t)$, then one can show

$$\sup_{t \geq 0} \mathbb{E} (|u(t, \cdot)|_\rho^2) < \infty$$

Proof. Fix an arbitrary $T > 0$ and denote $C_T := H(T, \gamma_2)^{1/2}$. For any $t \in [0, T]$, by Fubini's Theorem and the moment bound (1.13), we see that

$$\mathbb{E} (|u(t, \cdot)|_\rho^2) = \mathbb{E} \left(\int_{\mathbb{R}^d} |u(t, x)|^2 \rho(x) dx \right) \leq C_T \int_{\mathbb{R}^d} \left(1 + (|\varphi| * G(t, \cdot))(x) \right)^2 \rho(x) dx.$$

By definition $\rho \in L^1(\mathbb{R}^d)$, hence we see that

$$\mathbb{E} (|u(t, \cdot)|_\rho^2) \leq 2C_T \left[\|\rho\|_{L^1(\mathbb{R}^d)} + \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x-y) |\varphi(y)| dy \right)^2 \rho(x) dx \right]. \quad (2.12) \quad \boxed{\text{mstep}}$$

Another application of the Cauchy-Schwartz inequality shows that

$$\mathbb{E} (|u(t, \cdot)|_\rho^2) \leq 2C_T \left[\|\rho\|_{L^1(\mathbb{R}^d)} + \int_{\mathbb{R}^d} (\rho * G(t, \cdot))(y) |\varphi(y)|^2 dy \right].$$

Then an application of the admissible weight property (2.3) proves (2.11).

Hence,

$$\sup_{0 \leq t \leq t_0} \mathbb{E} |u(t, \cdot)|_\rho^2 < \infty.$$

Now consider $t \geq t_0 > 0$ and further assume that $\Upsilon(0) < \infty$ and that $\sup_{x \in \mathbb{R}^d} (|\varphi| * G(t, \cdot))(x) \leq K(t)$ for any $t > 0$. Now starting with (2.12), we see that

$$\mathbb{E} (|u(t, \cdot)|_\rho^2) \leq C \left[1 + \int_{\mathbb{R}^d} (G(t, \cdot) * |\varphi(\cdot)|)(x)^2 \rho(x) dx \right]$$

However,

$$(G(t, \cdot) * |\varphi(\cdot)|)(x) = (G(t - t_0, \cdot) * G(t_0, \cdot) * |\varphi(\cdot)|)(x) \leq K(t_0) (G(t - t_0, \cdot) * 1)(x) = K(t_0)$$

where $K(t_0)$ is the bound for $\sup_{x \in \mathbb{R}^d} (|\varphi| * G(t, \cdot))(x)$. Now continuing, we see that

$$\begin{aligned} \mathbb{E} (|u(t, \cdot)|_\rho^2) &\leq C \left[1 + \int_{\mathbb{R}^d} K(t_0)^2 \rho(x) dx \right] \\ &= C \left[1 + K(t_0)^2 \|\rho\|_{L^1(\mathbb{R}^d)} \right] < \infty. \end{aligned}$$

This proves that

$$\sup_{t \geq t_0} \mathbb{E} |u(t, \cdot)|_\rho^2 < \infty$$

thus completing the proof of the claim. $\square$

3 Proof of Theorems 1.6 and 1.9

We start by stating a lemma that will be used to complete the proof. The proof of the lemma is very technical and serves no real purpose in understanding its use to complete the proof of Theorem 1.6. For that reason we postpone proving it until later.

L:Main

Lemma 3.1. *Under the assumptions of Theorem 1.6 and for arbitrary $\epsilon \in (0, 1)$, $0 < t_0 \leq 1$ and $R > 0$ there exists a compact set $\mathcal{K}$ in L_ρ^2 such that*

$$\mathbb{P}\{u^\hat{\varphi}(t, \cdot) \in \mathcal{K}\} \geq (1 - \epsilon)\mathbb{P}\{|u^\hat{\varphi}(t - t_0, \cdot)|_{\hat{\rho}} < R\} \quad \text{for all } t > t_0. \quad (3.1)$$

E:cpt

We may now complete the proof of Theorem 1.6, but first we must recall two definitions.

D:bdinprob

Definition 3.2 (Bonded in Probability). The process $u^\varphi(t, x)$ is said to be *bounded in probability* in $L_\rho^2(\mathbb{R}^d)$ if

$$\forall \epsilon > 0, \exists R > 0 : \forall t > 0, \mathbb{P}\{|u^\varphi(t, \cdot)|_\rho \geq R\} < \epsilon. \quad (3.2)$$

D:tight

Definition 3.3 (Tightness of a family of measures). Consider the Hausdorff space (X, T) and suppose that the sigma algebra Σ contains T . Let M be a collection of probability measures defined on Σ . The collection M is said to be *tight* if for any $\epsilon > 0$, there exists a compact set $K_\epsilon \subset X$ such that for any measure $m \in M$, $m(K_\epsilon) > 1 - \epsilon$.

Proof of Theorem 1.6. First recall that by Theorem 2.8

$$\sup_{t \geq 0} \mathbb{E}|u^\hat{\varphi}(t, \cdot)|_\rho^2 < \infty. \quad (3.3)$$

E:bipc

In fact, this shows that $u^\hat{\varphi}(t, \cdot)$ is bonded in probability in $L_\rho^2(\mathbb{R}^d)$. To see this, let $\sup_{t \geq 0} \mathbb{E}|u(t, \cdot)|_\rho^2 = B$ and let $\lambda > 0$. Then by applying Chebyshev's inequality we may choose $C > 0$ large enough so that

$$\mathbb{P}[|u(t, \cdot)|_\rho \geq C] \leq C^{-2} \mathbb{E}|u(t, \cdot)|_\rho^2 \leq C^{-2} B < \lambda.$$

Hence $u(t, \cdot)$ is bounded in probability in L_ρ^2 .

Now let $\epsilon \in (0, 1)$. Since $u^\hat{\varphi}(t, x)$ is bounded in probability, we can choose $R > 0$ such that for any $t \geq t_0$,

$$\mathbb{P}\{|u^\hat{\varphi}(t - t_0, \cdot)|_{\hat{\rho}} \geq R\} < \epsilon.$$

By applying Lemma 3.1, we can find a compact set $\mathcal{K} \in L_\rho^2$ such that

$$\mathbb{P}\{u^\hat{\varphi}(t, \cdot) \in \mathcal{K}\} \geq (1 - \epsilon)\mathbb{P}\{|u^\hat{\varphi}(t - t_0, \cdot)|_{\hat{\rho}} < R\} \quad \text{for all } t > t_0.$$

In return, this implies that the family of laws $\mathcal{L}(u^\hat{\varphi}(t, \cdot))$ for $t \geq t_0$ is tight in L_ρ^2 . Indeed, if we multiply both sides of (3.1) by -1 and then add 1 to both sides we see that

$$\begin{aligned} \mathbb{P}\{u^\hat{\varphi}(t, \cdot) \notin \mathcal{K}\} &\leq \mathbb{P}\{|u^\hat{\varphi}(t - t_0, \cdot)|_{\hat{\rho}} \geq R\} + \epsilon \mathbb{P}\{|u^\hat{\varphi}(t - t_0, \cdot)|_{\hat{\rho}} < R\} \\ &\leq 2\epsilon. \end{aligned}$$

Therefore, the family of probability measures

$$\frac{1}{T} \int_{t_0}^{T+t_0} \mathcal{L}(u^\hat{\varphi}(t, \cdot)) dt, \quad T > 0 \quad (3.4)$$

E:intlaw

is tight as well; see [DZ96, Corollary 3.1.2]. We may say that (3.4) has a subsequence that weakly converges to some measure η and that in fact η is invariant for the laws of (2.6). $\square$

We now prove a lemma which will be used to prove Theorem 1.9.

L:Moments

Lemma 3.4. *Consider the stochastic heat equation (1.1) and suppose the assumption of Theorem 1.6 remain true and (1.17) holds true. Then for any $t_0 > 0$, $u(t_0, \cdot) \in L_\rho^2$ almost surely for any admissible weight. Also the solution to the following stochastic heat equation*

$$\begin{cases} \frac{\partial v}{\partial t}(t, x) - \frac{1}{2} \Delta v(t, x) = b(x, v(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ v(0, x) = u(t_0, x) \end{cases} \quad (3.5)$$

E:vspde2

satisfies the property that

$$\sup_{t \geq 0} \mathbb{E} |v(t, \cdot)|_\rho^2 < \infty.$$

Proof. We first show that $u(t_0, x) \in L_\rho^2$ for any admissible weight.

$$\begin{aligned} \mathbb{E} |u(t_0, \cdot)|_\rho^2 &= \int_{\mathbb{R}^d} dx \|u(t_0, x)\|_2^2 \rho(x) \\ &\leq C \int_{\mathbb{R}^d} dx (1 + (|\mu| * G(t_0, \cdot)(x)))^2 \rho(x) \end{aligned}$$

which follows from the moment estimates (1.15). Now by applying (1.17) we see that

$$\begin{aligned} \mathbb{E} |u(t_0, \cdot)|_\rho^2 &\leq C \int_{\mathbb{R}^d} dx (1 + K(t_0))^2 \rho(x) \\ &= C (1 + K(t_0))^2 \|\rho\|_{L^1(\mathbb{R}^d)} \end{aligned}$$

which shows that $u(t_0, x) \in L_\rho^2$ almost surely for any admissible weight. Now it remains to show $\sup_{t \geq 0} \mathbb{E} |v(t, \cdot)|_\rho^2 < \infty$.

$$\begin{aligned} \mathbb{E} |v(t, \cdot)|_\rho^2 &= \int_{\mathbb{R}^d} \|v(t, x)\|_2^2 \rho(x) dx \\ &\leq C \int_{\mathbb{R}^d} (1 + (\|u(t_0, \cdot)\|_2 * G(t, \cdot)(x)))^2 \rho(x) dx \end{aligned}$$

which follows from an extension of moment estimates (1.15) for random initial conditions. We now continue by again applying the moment estimates.

$$\begin{aligned} \mathbb{E} |v(t, \cdot)|_\rho^2 &\leq C \int_{\mathbb{R}^d} \left(1 + ([1 + (\mu * G(t_0, \cdot))](\cdot) * G(t, \cdot))(x) \right)^2 \rho(x) dx \\ &= C \int_{\mathbb{R}^d} \left(1 + \left(1 + [(\mu * G(t_0, \cdot))](\cdot) * G(t, \cdot) \right)(x) \right)^2 \rho(x) dx \end{aligned}$$

Now an application of (1.17) gives us

$$\begin{aligned} \mathbb{E} |v(t, \cdot)|_\rho^2 &\leq C \int_{\mathbb{R}^d} \left(1 + \left(2 + K(t_0) \right) \right)^2 \rho(x) dx \\ &= C \left(3 + K(t_0) \right)^2 \|\rho\|_{L^1 \mathbb{R}^d} \end{aligned}$$

which is true for every $t > 0$. This shows $\sup_{t \geq 0} \mathbb{E} |v(t, \cdot)|_\rho^2 < \infty$. $\square$

We now can prove Theorem 1.9.

Proof of Theorem 1.9. Pick $t_0 > 0$ and $l_0 > 0$ such that $l = t_0 + l_0$ for a given $l > 0$. By Theorem 1.6, there exists an invariant measure, η , for $\{\mathcal{L}(v(t, \cdot))\}_{t \geq l_0}$ where v is the solution to (3.5). However, by a general property of the heat solution, $v(t, x) = u^\mu(t + t_0, x)$ and so η is invariant for $\{\mathcal{L}(u^\mu(t + t_0, \cdot))\}_{t \geq l_0} = \{\mathcal{L}(u^\mu(t, \cdot))\}_{t \geq l}$. $\square$

4 Proof of Lemma 3.1

We start off with a proposition whose proof is given in Proposition 2.1 [TZ98]

P:weight

Proposition 4.1. *For any admissible weight ρ , the operators on $L_\rho^2(\mathbb{R}^d)$ defined by $\varphi \mapsto (G(t, \cdot) * \varphi(\cdot))(x)$ can be extended to a C_0 -semigroup on L_ρ^2 . Moreover, if $\widehat{\rho}$ is another admissible weight such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\widehat{\rho}(x)} dx < \infty$$

then for any $t < 0$, the operators defined above are compact from $L_\rho^2(\mathbb{R}^d)$ to $L_{\widehat{\rho}}^2(\mathbb{R}^d)$.

We now define two operators. Let $q > 2$ and $t_0 > 0$ be arbitrary and $\alpha \in (q^{-1}, 2^{-1})$. Define the operator $F_\alpha^{t_0}$ as follows

$$(F_\alpha^{t_0} f)(x) := \int_0^{t_0} ds \int_{\mathbb{R}^d} dy (t_0 - s)^{\alpha-1} G(t_0 - s, x - y) f(s, y). \quad (4.1)$$

E:for

It is shown in the proof of Theorem 3.1 [TZ98] that F_α is compact from $L^q((0, t_0), L_\rho^2)$ to L_ρ^2 where $L^q((0, t_0), L_\rho^2)$ is the space of functions that map $s \mapsto f(s, \cdot)$ and $f(s, \cdot) \in L_\rho^2(\mathbb{R}^d)$ and in addition $\int_0^{t_0} \|f(s, \cdot)\|_\rho^q ds < \infty$.

Next we define for $\varphi \in L_\rho^2(\mathbb{R}^d)$ the operators $Y^\varphi(s, y)$, or simply as $Y(s, y)$ when there is no confusion, as follows

$$Y^\varphi(s, y) = \int_0^s \int_{\mathbb{R}^d} (s - r)^{-\alpha} G(s - r, y - z) b(u^\varphi(r, z)) W(dr, dz). \quad (4.2)$$

E:yop

L:yineq

Lemma 4.2. *Let $q > 2$, then Y^φ operators, (4.2), satisfies the inequality*

$$\mathbb{E} \int_0^1 |Y^\varphi(s, \cdot)|_\rho^q ds \leq K |\varphi|_\rho^q. \quad (4.3)$$

E:yineq

Proof. First, notice by applying Minkowski's Integral Inequality that

$$\begin{aligned} E|Y(s, \cdot)|_\rho^q &= \int_\Omega dP \left(\int_{\mathbb{R}^d} dy Y(s, y)^2 \rho(y) \right)^{q/2} \\ &\leq \left(\int_{\mathbb{R}^d} dy \left(\int_\Omega dP Y(s, y)^q \rho(y)^{q/2} \right)^{2/q} \right)^{q/2} \\ &= \left(\int_{\mathbb{R}^d} dy \rho(y) \|Y(s, y)\|_q^2 \right)^{q/2} \\ &= \|Y(s, \cdot)\|_q^q \end{aligned}$$

and because of this, we look to find an appropriate upper bound for $\|Y(s, y)\|_q^2$ which for now we denote as $M(s, y)$. Then we integrate this bound

$$\int_{\mathbb{R}^d} M(s, y) \rho(y) dy$$

which gives us a bound for $\|Y(s, y)\|_q$. To do this, we start by applying the Burkholder-Davis-Gundy inequality, Theorem 5.27 [DK+09], as well as Minkowski's integral inequality to see that

$$\|Y(s, y)\|_q^2 \leq C_q \int_0^s dr (s - r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) \|b(u(r, z_1))\|_q f(z_1 - z_2)$$

$$\times G(s-r, y-z_2) \|b(u(r, z_2))\|_q$$

where C_q is the standard constant from the BDG-inequality. Note that for Lipschitz functions, we have that $|b(u)| < k(1 + |u|)$. We apply this and the moment bound (1.15) to $\|b(u(r, z_i))\|_q$ above to see that

$$\begin{aligned} \|b(u(r, z_i))\|_q &\leq k(1 + \|u(r, z_i)\|_q) \\ &\leq k \left(1 + (1 + J_0(r, z_i)) \right) \quad \text{by (1.15)} \\ &\leq k(1 + J_0(r, z_i)). \end{aligned}$$

Now applying this above to the inequality for $\|Y(s, y)\|_q^2$ we get that

$$\begin{aligned} \|Y(s, y)\|_q^2 &\leq k \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G(s-r, y-z_i) (1 + J_0(r, z_i)) \\ &= k \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) \\ &\quad + k \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) J_0(r, z_1) \\ &\quad + k \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) J_0(r, z_2) \\ &\quad + k \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) J_0(r, z_1) J_0(r, z_2) \\ &=: I_1 + I_2 + I_3 + I_4. \end{aligned}$$

We now investigate each of the integrals I_k above. Throughout the remainder of this proof and the proof of Lemma 4.2 we will be using the following relation

$$G(s-r, y-z_i) G(r, z_i - \sigma) = G(s, y - \sigma) G\left(\frac{r(s-r)}{s}, z_i - \sigma - \frac{r}{s}(y - \sigma)\right). \quad (4.4) \quad \boxed{\text{E:gsim}}$$

For I_1 :

$$\begin{aligned} I_1 &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1 - z_2) G(s-r, y-z_2) \\ &= \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \Gamma(d\xi) (G(s-r, y - \cdot) * G(s-r, y + \cdot))(\xi). \end{aligned}$$

Now recalling that Γ has density f and by applying Plancherel's theorem, I_1 is bounded above by

$$(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2).$$

Now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ we get that

$$\begin{aligned} \int_{\mathbb{R}^d} I_1 \cdot \rho(y) dy &\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2) \int_{\mathbb{R}^d} \rho(y) dy \\ &\leq (2\pi)^{-2d} \cdot \|\rho\|_{L^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2). \end{aligned}$$

For I_2 :

$$\begin{aligned}
I_2 &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) J_0(r, z_1) \\
&= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) \int_{\mathbb{R}^d} d\sigma G(r, z_1-\sigma) |\varphi(\sigma)| \\
&= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_2) f(z_1-z_2) G\left(\frac{r(s-r)}{s}, z_1-\sigma-\frac{r}{s}(y-\sigma)\right) \\
&\quad \times \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)| \\
&\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|.
\end{aligned}$$

Now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ and apply the admissible weight property we get that

$$\begin{aligned}
\int_{\mathbb{R}^d} I_2 \cdot \rho(y) dy &\leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \\
&\quad \times \int_{\mathbb{R}^d} dy \rho(y) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)| \\
&\leq (2\pi)^{-2d} C(S) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \\
&\quad \times \int_{\mathbb{R}^d} d\sigma \rho(\sigma) |\varphi(\sigma)| \\
&= (2\pi)^{-2d} C(S) \|\varphi\|_{L^1_\rho(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right)
\end{aligned}$$

where the above is true for any $0 \leq s \leq S$, $S > 0$ is arbitrary and $C(S)$ is the constant from the admissible weight property.

For I_3 : By completing the same steps as for I_2 we get that

$$I_3 \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|$$

and now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ we will get the same bound as in for I_2 above. Namely,

$$\int_{\mathbb{R}^d} I_3 \cdot \rho(y) dy \leq (2\pi)^{-2d} C(S) \|\varphi\|_{L^1_\rho(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right).$$

For I_4 :

$$\begin{aligned}
I_4 &= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) J_0(r, z_1) J_0(r, z_2) \\
&= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) \\
&\quad \times \iint_{\mathbb{R}^{2d}} d\sigma_1 d\sigma_2 G(r, z_1-\sigma_1) G(r, z_2-\sigma_2) |\varphi(\sigma_1)| |\varphi(\sigma_2)| \\
&= \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} d\sigma_1 d\sigma_2 G(s, y-\sigma_1) G(s, y-\sigma_2) |\varphi(\sigma_1)| |\varphi(\sigma_2)|
\end{aligned}$$

$$\begin{aligned}
& \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G\left(\frac{r(s-r)}{s}, z_i - \sigma_i - \frac{r}{s}(y - \sigma_i)\right) \\
& \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} d\sigma_1 d\sigma_2 G(s, y - \sigma_1) G(s, y - \sigma_2) |\varphi(\sigma_1)| |\varphi(\sigma_2)| \\
& \quad \times \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\
& = (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \left(\int_{\mathbb{R}^d} d\sigma G(s, y - \sigma) |\varphi(\sigma)|\right)^2 \\
& \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y - \sigma) |\varphi(\sigma)|^2.
\end{aligned}$$

Now if we integrate this in y over $\mathbb{R}^d$ with respect to $\rho(y)dy$ we get that

$$\begin{aligned}
\int_{\mathbb{R}^d} I_4 \cdot \rho(y) dy & \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\
& \quad \times \int_{\mathbb{R}^d} dy \rho(y) \int_{\mathbb{R}^d} d\sigma G(s, y - \sigma) |\varphi(\sigma)|^2 \\
& \leq (2\pi)^{-2d} C(S) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \int_{\mathbb{R}^d} d\sigma \rho(\sigma) |\varphi(\sigma)|^2 \\
& = (2\pi)^{-2d} C(S) |\varphi|_\rho^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right).
\end{aligned}$$

Now, putting all of this together, we see that

$$\begin{aligned}
(E|Y(s, y)|_\rho^q)^{2/q} & \leq \int_{\mathbb{R}^d} dy \rho(y) \|Y(s, y)\|_q^2 \\
& \leq \int_{\mathbb{R}^d} dy \rho(y) (I_1 + I_2 + I_3 + I_4) \\
& \leq (2\pi)^{-2d} \|\rho\|_{L^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\
& \quad + 2(2\pi)^{-2d} C(S) \|\varphi\|_{L_\rho^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right) |\xi|^2\right) \\
& \quad + (2\pi)^{-2d} C(S) |\varphi|_\rho^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right)
\end{aligned}$$

and by assumption 1.9 and Lemma 4.2 below, the three integrals above are all finite any $s \geq 0$ so we may take the maximum for $0 \leq s \leq S$ and say there exists some constant K_S dependent on S such that

$$E|Y(s, \cdot)|_\rho^q \leq K_S |\varphi|_\rho^q, \quad \forall s \in [0, S].$$

Lastly if we integrate out s from $0 \rightarrow 1$ we get our desired inequality,

$$\int_0^1 E|Y(s, \cdot)|_\rho^q ds \leq \int_0^1 K_1 |\varphi|_\rho^q ds = K_1 |\varphi|_\rho^q.$$

□

L:equiv

Lemma 4.3. Assuming (1.9) then we may say the following two integrals are finite

$$\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \quad (4.5) \quad \text{E:fcon2}$$

and

$$\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp \left[- \left(\frac{s-r}{2} + \frac{r(s-r)}{2s} \right) |\xi|^2 \right]. \quad (4.6) \quad \boxed{\text{E: fcon3}}$$

Proof. We first focus on (4.5). By applying the substitution $u = s - r$ the we may rewrite

$$(1.9) = \int_0^s du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-u|\xi|^2)$$

and

$$(4.5) = \int_0^s du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp \left(- \frac{u(s-u)}{s} |\xi|^2 \right).$$

Since $u(s-u) \geq \frac{us}{2}$ for all $u \in [0, s]$ then we may say that

$$\begin{aligned} (4.5) &\leq \int_0^s du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp \left(- \frac{u}{2} |\xi|^2 \right) \\ &= 2^{1-2\alpha} \int_0^{s/2} dt t^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-t|\xi|^2) \quad \text{substitution } t = \frac{u}{2} \\ &\leq (1.9) < \infty \end{aligned}$$

which completes the proof showing assumption (1.9) implies that (4.5) is finite. The finiteness of (4.6) follows easily by noting that

$$\begin{aligned} (4.6) &\leq \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp \left(- \frac{s-r}{2} |\xi|^2 \right) \\ &= 2^{1-2\alpha} \int_0^{s/2} du u^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-u|\xi|^2) \quad \text{using the substitution } u = \frac{s-r}{2} \\ &\leq (1.9) < \infty \end{aligned}$$

which completes the proof showing that assumption (1.9) implies (4.6). $\square$

Lemma 4.4. *The following factorization holds for $\alpha \in (0, 1)$*

$$\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds = \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(u(r, z)) W(dr, dz). \quad (4.7) \quad \boxed{\text{E: fac}}$$

Proof.

$$\begin{aligned} &\frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G(t-s, \cdot) * Y(s, \cdot)](x) ds \\ &\stackrel{1}{=} \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(u(r, z)) W(dr, dz) \\ &\stackrel{2}{=} \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(u(r, z)) W(dr, dz) \\ &\stackrel{3}{=} \frac{\sin(\alpha\pi)}{\pi} \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(u(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \\ &\stackrel{4}{=} \int_0^t \int_{\mathbb{R}^d} G(t-r, x-z) b(u(r, z)) W(dr, dz) \end{aligned}$$

where the last step is true provided $\alpha \in (0, 1)$. $\square$

Remark 4.5. We also need to verify the use of Fubini's Theorem which is stated in stated in [Wal86]. We used Fubini's Theorem twice above: first from equality 1 to 2 and then from equality 3 to 4. For reference, we will state Fubini's Theorem as given in [Wal86] and verify the two uses of it in the proof of Lemma 4.2.

Theorem 4.6 (Fubini's Theorem). *Let $(G, \mathcal{G}, \mu)$ be a finite measure space and $(E, \mathcal{E}, \nu)$ be a σ -finite measure space. Let $g(r, z, \omega, y)$, $z \in E$, $r \geq 0$, $\omega \in \Omega$, $y \in G$ be a $\mathcal{E} \times \mathcal{G}$ measurable function. Suppose that*

$$\mathbb{E} \left[\int_{E \times E \times [0, T] \times G} |g(r, z_1, \omega, y) g(r, z_2, \omega, y)| K(dz_1, dz_2, dr) \mu(dy) \right] < \infty \quad (4.8) \quad \boxed{\text{E:fcond}}$$

where K is the dominating measure as in [DK⁺09]. Then

$$\int_G \left[\int_E \int_0^s g(r, z, y) W(dr, dz) \right] \mu(dy) = \int_E \int_0^s \left[\int_G g(r, z, y) \mu(dy) \right] W(dr, dz).$$

Remark 4.7. Following [DK⁺09], the dominating measure for this problem is given by

$$K(dz_1, dz_2, dr) = f(z_1 - z_2) dz_1 dz_2 dr$$

where f is the correlation function of the spatially homogeneous Gaussian noise.

We now verify (4.8) for both applications of Fubini's Theorem.

From Equality 1 to 2:

Proof. We let

$$\begin{aligned} g(z, r, \omega, y) &= (s - r)^{-\alpha} G(s - r, y - z) b(u(r, z)) \\ (G, \mathcal{G}, \mu) &= (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), G(t - s, x - y) dy) \\ (E, \mathcal{E}, \nu) &= (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda) \end{aligned}$$

and so (4.8) becomes

$$\begin{aligned} (4.8) &= \mathbb{E} \int_{\mathbb{R}^d} dy G(t - s, x - y) \int_0^s dr (s - r)^{-2\alpha} \\ &\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) |b(u(r, z_1))| f(z_1 - z_2) G(s - r, y - z_2) |b(u(r, z_2))|. \end{aligned}$$

For simplicity, we first investigate the $\mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2$ part of the integral. We continue by applying the Lipschitz property of b .

$$\begin{aligned} &\mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) |b(u(r, z_1))| f(z_1 - z_2) G(s - r, y - z_2) |b(u(r, z_2))| \\ &\leq k_1 \mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) (1 + u(r, z_1)) f(z_1 - z_2) G(s - r, y - z_2) (1 + u(r, z_2)) \\ &\leq k_1 \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) f(z_1 - z_2) G(s - r, y - z_2) \\ &\quad + k_1 \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s - r, y - z_1) \|u(r, z_1)\|_1 f(z_1 - z_2) G(s - r, y - z_2) \end{aligned}$$

$$\begin{aligned}
& + k_1 \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) f(z_1-z_2) G(s-r, y-z_2) \|u(r, z_2)\|_1 \\
& + k_1 \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|u(r, z_1)\|_2 f(z_1-z_2) G(s-r, y-z_2) \|u(r, z_2)\|_2 \\
& =: L_1 + \dots + L_4.
\end{aligned}$$

For L_1 : If we back at I_1 from the proof of Lemma 4.2 then we see that

$$L_1 \leq (2\pi)^{-2d} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2)$$

and now if we include the $\int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha}$ part of the integral we get that

$$\int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} L_1 \leq (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2).$$

For L_2 : We first use the moment bounds from [CH19] along with the fact that $\|u(r, z_1)\|_1 \leq \|u(r, z_1)\|_2$.

$$\begin{aligned}
L_2 & \leq k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) (1 + J_0(r, z_1)) f(z_1-z_2) G(s-r, y-z_2) \\
& = kL_1 + \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) J_0(r, z_1) f(z_1-z_2) G(s-r, y-z_2).
\end{aligned}$$

But this last integral above is the same thing as I_2 from Corollary 4.1 so we see that

$$L_2 \leq kL_1 + (2\pi)^{-2d} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)|$$

and now if we include the $\int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha}$ part of the integral we get that

$$\begin{aligned}
& \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} L_2 \\
& \leq k \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} L_1 \\
& \quad + (2\pi)^{-2d} \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \\
& \quad \quad \quad \times \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)| \\
& \leq k(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\
& \quad + (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \\
& \quad \quad \quad \times \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma) |\varphi(\sigma)| \\
& = k(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\
& \quad + (2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)| \\
& \leq k(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2)
\end{aligned}$$

$$+ (2\pi)^{-2d} \|G(t, x - \cdot) \varphi(\cdot)\|_{L^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right).$$

For L_3 : Following the same steps for L_2 will get you the same exact bound as for L_3 .

For L_4 : We continue as we did for L_2 to see that

$$\begin{aligned} L_4 &\leq k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)(1+J_0(r, z_1))f(z_1-z_2)G(s-r, y-z_2)(1+J_0(r, z_2)) \\ &= k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)f(z_1-z_2)G(s-r, y-z_2) \\ &\quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)J_0(r, z_1)f(z_1-z_2)G(s-r, y-z_2) \\ &\quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)f(z_1-z_2)G(s-r, y-z_2)J_0(r, z_2) \\ &\quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)J_0(r, z_1)f(z_1-z_2)G(s-r, y-z_2)J_0(r, z_2). \end{aligned}$$

These first three integrals have already been handled above in L_1, L_2 and L_3 of this proof so we only need to proceed with the fourth integral. However, notice from I_4 above in the proof of Lemma 4.2 that this fourth integral is bounded by the following

$$\begin{aligned} &\iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1)J_0(r, z_1)f(z_1-z_2)G(s-r, y-z_2)J_0(r, z_2) \\ &\leq (2\pi)^{-2d} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma)|\varphi(\sigma)|^2 \end{aligned}$$

and now if we include the $\int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha}$ part of the integral we get that

$$\begin{aligned} &\int_{\mathbb{R}^d} dy G(t-s, x-y) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma)|\varphi(\sigma)|^2 \\ &= \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \int_{\mathbb{R}^d} dy G(t-s, x-y) \int_{\mathbb{R}^d} d\sigma G(s, y-\sigma)|\varphi(\sigma)|^2 \\ &= \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \int_{\mathbb{R}^d} d\sigma G(t, x-\sigma)|\varphi(\sigma)|^2 \\ &\leq \|G(t, x - \cdot) \varphi(\cdot)\|_{L^1(\mathbb{R}^d)}^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) \end{aligned}$$

and now putting this all together for L_4 we see that

$$\begin{aligned} &\int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} L_4 \\ &\leq k(2\pi)^{-2d} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2) \\ &\quad + 2k(2\pi)^{-2d} \|G(t, x - \cdot) \varphi(\cdot)\|_{L^1(\mathbb{R}^d)} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \\ &\quad + k(2\pi)^{-2d} \|G(t, x - \cdot) \varphi(\cdot)\|_{L^1(\mathbb{R}^d)}^2 \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right). \end{aligned}$$

Considering the above arguments we may say that there is a constant $K_{x,t}$ depending on x and t such that

$$(4.8) \leq K_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(s-r)|\xi|^2)$$

$$\begin{aligned}
& + K_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)\right) \\
& + K_{x,t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) < \infty
\end{aligned}$$

where the finiteness follows by (1.9) and Lemma 4.3. This completes the justification for the first use of Fubini's Theorem. $\square$

From Equality 2 to 3:

Proof. We let

$$\begin{aligned}
g(r, z, \omega, y) &= (s-r)^{-\alpha} G(t-r, x-z) b(u(r, z)) \\
(G, \mathcal{G}, \mu) &= ((0, t), \mathcal{B}(0, t), (t-s)^{\alpha-1} ds) \\
(E, \mathcal{E}, \nu) &= (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)
\end{aligned}$$

and so (4.8) becomes

$$(4.8) = \mathbb{E} \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G(t-r, x-z_i) |b(u(r, z_i))|$$

and again for simplicity we will first investigate the $\mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2$ integral. Using the Lipschitz property, we similarly get that

$$\begin{aligned}
& \mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) |b(u(r, z_1))| f(z_1 - z_2) G(t-r, x-z_2) |b(u(r, z_2))| \\
& \leq k \mathbb{E} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) (1+u(r, z_1)) f(z_1 - z_2) G(t-r, x-z_2) (1+u(r, z_2)) \\
& \leq k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) f(z_1 - z_2) G(t-r, x-z_2) \\
& \quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) \|u(r, z_1)\|_1 f(z_1 - z_2) G(t-r, x-z_2) \\
& \quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) f(z_1 - z_2) G(t-r, x-z_2) \|u(r, z_2)\|_1 \\
& \quad + k \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) \|u(r, z_1)\|_2 f(z_1 - z_2) G(t-r, x-z_2) \|u(r, z_2)\|_2 \\
& =: K_1 + \dots + K_4.
\end{aligned}$$

We apply similar techniques as in for the first justification of Fubini's Theorem to get that

For K_1 :

$$\begin{aligned}
\int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} K_1 & \leq \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(t-r)|\xi|^2) \\
& \leq k_{t,1} \int_0^t ds (t-s)^{\alpha-1}
\end{aligned}$$

where

$$k_{t,1} = \max_{0 \leq s \leq t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(t-r)|\xi|^2)$$

and in fact $k_{t,1}$ is finite for each $t > 0$ by Lemma 4.3.

For K_2 and K_3 :

$$\begin{aligned}
& \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} K_2 \\
& \leq k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(t-r)|\xi|^2) \\
& \quad + k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{t-r}{2} + \frac{r(t-r)}{2t}\right)|\xi|^2\right) \\
& \quad \times \int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)| \\
& \leq (k_{t,1} + kJ_0(t, x)k_{t,2}) \int_0^t ds (t-s)^{\alpha-1}
\end{aligned}$$

where

$$k_{t,2} = \max_{0 \leq s \leq t} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{t-r}{2} + \frac{r(t-r)}{2t}\right)|\xi|^2\right)$$

which is finite by Lemma 4.3.

For K_4 :

$$\begin{aligned}
& \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} K_4 \\
& \leq k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp(-(t-r)|\xi|^2) | \\
& \quad + 2k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\left(\frac{t-r}{2} + \frac{r(t-r)}{2t}\right)|\xi|^2\right) \\
& \quad \times \int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)| \\
& \quad + k \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t}|\xi|^2\right) \left(\int_{\mathbb{R}^d} d\sigma G(t, x-\sigma) |\varphi(\sigma)|\right)^2 \\
& \leq k(k_{t,1} + 2k_{t,2}J_0(t, x) + k_{t,3}J_0(t, x)^2) \int_0^t ds (t-s)^{\alpha-1}
\end{aligned}$$

where

$$k_{t,3} = \max_{0 \leq s \leq t} \int_{\mathbb{R}^d} \widehat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t}|\xi|^2\right)$$

which is again finite by Lemma 4.3. Now we may say that there exists some constant K_t dependent on t such that (4.8) has the following bound which is finite since $0 < \alpha < 1$

$$(4.8) \leq K_t \left(1 + 2J_0(t, x) + J_0(t, x)^2\right) \int_0^t ds (t-s)^{\alpha-1}.$$

□

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Corollary 4.8. *Let $u(t, x)$ denote the solution to (2.5) and $\alpha \in (0, 1)$. Then for $t_0 > 0$ we may say that*

$$u(t_0, x) = (G(t_0, \cdot) * \varphi(\cdot))(x) + \frac{\sin(\alpha\pi)}{\pi} F_\alpha^{t_0}(Y(s, \cdot))(x). \quad (4.9)$$

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Proof. Follows directly from (4.7). □

We are now able to prove Lemma 3.1.

Proof of Lemma 3.1. We first define a set $\mathcal{K}(\Theta)$ as

$$\mathcal{K}(\Theta) = \{(G(\delta_0, \cdot) * y(\cdot))(x) + (F_\alpha h)(x), \quad \text{where } |y|_{\hat{\rho}} \leq \Theta \text{ and } |h|_{L^q(0,1,L^2_{\hat{\rho}})} \leq \Theta\}.$$

Then $\mathcal{K}(\Theta)$ is relatively compact in $L^2_{\hat{\rho}}$ for $\Theta > 0$. Moreover, by (4.3), (4.9) and Chebychev's inequality, and if we choose R and Θ so that $|\hat{\varphi}|_{\hat{\rho}} < R < \Theta$ then we have

$$\begin{aligned} \mathbb{P}\{u^{\hat{\varphi}}(\delta_0, \cdot) \notin \mathcal{K}(\Theta)\} &\leq \mathbb{P}\left[\left(\int_0^1 |Y^{\hat{\varphi}}(s, \cdot)|_{\hat{\rho}}^q ds\right)^{1/q} > \frac{\pi\Theta}{\sin(\alpha\pi)}\right] \\ &\leq \frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \mathbb{E} \int_0^1 |Y^{\hat{\varphi}}(s, \cdot)|_{\hat{\rho}}^q ds \\ &\leq \frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \left(1 + |\hat{\varphi}|_{\hat{\rho}}^q\right) \end{aligned}$$

which implies that

$$\{u^{\hat{\varphi}}(\delta_0, \cdot) \in \mathcal{K}(\Theta)\} \geq 1 - \frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \left(1 + |\hat{\varphi}|_{\hat{\rho}}^q\right) \geq 1 - \frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \left(1 + R^q\right).$$

Therefore, if we choose $\Theta > 0$ big enough such that $\frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \left(1 + R^q\right) < \epsilon$ then we get that

$$\mathbb{P}\{u^{\hat{\varphi}}(\delta_0, \cdot) \in \mathcal{K}(\Theta)\} \geq 1 - \epsilon.$$

We can now deduce from this and the semigroup property of the solution to the heat equation that

$$\mathbb{P}\{u^{\hat{\varphi}}(t, \cdot) \in \mathcal{K}(\Theta)\} \geq (1 - \epsilon) \mathbb{P}\{|u^{\hat{\varphi}}(t - \delta_0, \cdot)|_{\hat{\rho}} < R\}.$$

Since we assumed that $u^{\hat{\varphi}}$ is bounded in probability in $L^2_{\hat{\rho}}$ then for and $\epsilon > 0$ we can fix $R > 0$ such that $\mathbb{P}\{|u^{\hat{\varphi}}(t, \cdot)|_{\hat{\rho}} \geq R\} < \epsilon$ and then taking $\Theta > R$ such that $\frac{\sin^q(\alpha\pi)}{\pi^q\Theta^q} \left(1 + R^q\right) < \epsilon$ concludes the claim. □

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